

1. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the profit percentage earned on selling five different articles P, Q, R, S and T. It also shows the discount allowed (in Rs.) on each article.

Articles	The profit percentage	The discount allowed (in Rs.)
P	20%	30
Q	40%	40
R	10%	54
S	25%	68
T	12.50%	50

Note:

The profit should be calculated on the cost price unless mentioned in the question.

The cost price of article P is half of article Q.

The selling price of article Q is Rs. 130 more than the marked price of article P.

The difference between the cost price of article R and S is Rs. 140. ($R < S$)

The marked price of article R is Rs. 148 less than the selling price of article S.

Article T has been marked up by 25% above its cost price.

If there is another article U, whose cost price is 50% more than that of article Q, and article U is marked up by [Y]% above its cost price, then sold at a profit of 33.33%, the discount allowed on article U is ₹80. Find the value of [Y/12].

- a. 1
- b. 4
- c. 5
- d. 6
- e. 8

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2. Questions

Find the difference between the mark-up percentage of articles Q and S.

- a. 9.25
- b. 20.50
- c. 6.50
- d. 13.75

e. 15.50

3. Questions

If the cost price of article R is increased by Rs. 60 but the mark-up percentage remained unchanged and sold a discount of Rs. 36, then find the ratio of the original selling price and new selling price of article R.

- a. 33:50
- b. 21:49
- c. 18:37
- d. 40:11
- e. 55:23

4. Questions

Find the selling price of article P is what percent of the marked price of article T?

- a. 36%
- b. 18%
- c. 40%
- d. 10%
- e. 24%

5. Questions

If there is an additional discount of $[D]\%$ is allowed on article P, while the cost price and marked price remained unchanged, then the profit earned on selling article P will be Rs. $[D-2]$. The value of $[D]$ must be between $__[M]__$ and $__[N]__$.

Which of the following does satisfy the value of $[M]$ and $[N]$?

I: $M = 8$ and $N = 12$

II: $M = 5$ and $N = 9$

III: $M = 6$ and $N = 15$

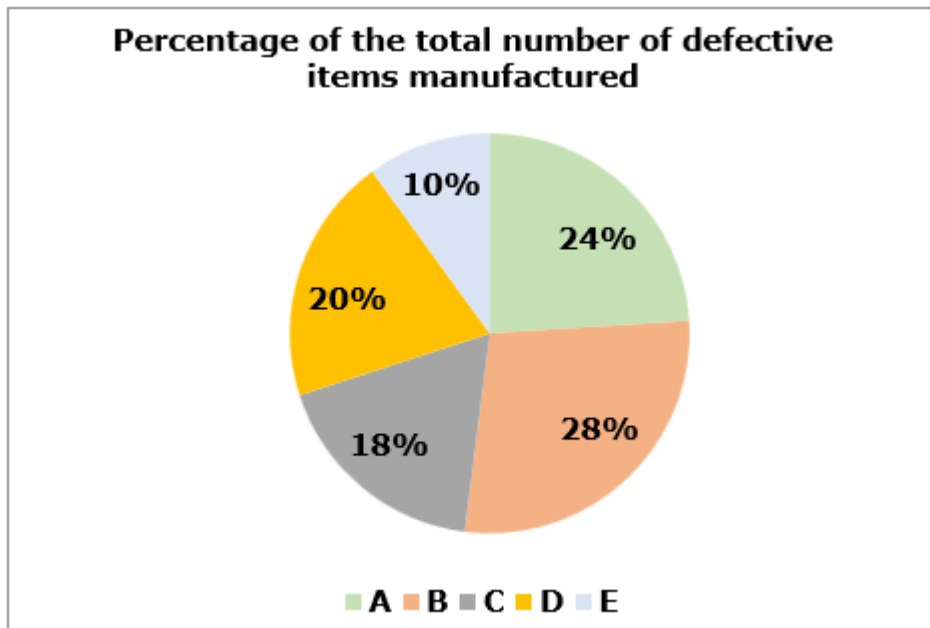
- a. Only I follow
- b. Both I and II follow
- c. Both I and III follow
- d. Only III follow
- e. All I, II and III follow

6. Questions

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Read the following information carefully and answer the questions based on it.

The pie chart given below shows the percentage distribution of the total number of defective items manufactured by five shops A, B, C, D, and E.



Note:

The total number of items manufactured = the number of defective items manufactured + the number of non-defective items manufactured

The total number of non-defective items manufactured = the number of non-defective items sold + the number of non-defective items unsold

Only non-defective items can be sold.

The total number of items manufactured (defective + non-defective) by shops A, B, C, D, and E is 120, 112, 108, 80, and 50 respectively.

The number of non-defective items sold by shop A is 72, which is 44 more than the number of non-defective items unsold by shop B.

The ratio of the non-defective items unsold by shop A and the number of non-defective items sold by shop B is 3:7.

The number of non-defective items sold by shop C is 6 less than that of shop B.

The number of non-defective items unsold by shop A and D are same.

In shop E, the difference between the number of non-defective items sold and unsold is 24. (sold > unsold)

If the ratio of the price of each item in shop D and E is 4:3 and the total revenue generated from the sale of all non-defective items sold by shop D is Rs. 4800 more than shop E, then find the difference between the price of each item in shop D and E.

- Rs. 200
- Rs. 160

- c. Rs. 320
- d. Rs. 280
- e. Rs. 100

7. Questions

In which of the following shop, the percentage of the number of non-defective items unsold with respect to the total number of non-defective items manufactured is highest?

- a. Shop A
- b. Shop B
- c. Shop C
- d. Shop D
- e. Shop E

8. Questions

Find the ratio of the number of non-defective items sold by shop C and the number of non-defective items unsold by shop A.

- a. 21:19
- b. 25:12
- c. 32:7
- d. 37:5
- e. 41:23

9. Questions

In shop A, if the number of defective wooden items manufactured is 6, which is $\frac{1}{6}^{\text{th}}$ of the number of non-defective wooden items manufactured, then find the ratio of the number of defective and non-defective steel items manufactured by shop A. (consider wooden and steel items only manufactured)

- a. 4:11
- b. 5:9
- c. 2:7
- d. 3:10
- e. 6:13

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10. Questions

If 40% of the total number of non-defective items sold by shop C are sold to females, which is

equal to the number of non-defective items sold to females by shop B, then find the total number of non-defective items sold to males by shop B and C together.

- a. 66
- b. 40
- c. 72
- d. 32
- e. 84

11. Questions

The following question has two quantities as Quantity I and Quantity II. You have to determine the relationship between them and give an answer accordingly.

Quantity I: Find the initial quantity of water in the vessel. A vessel contains mixture of milk and water in the ratio of 3:2. When 25% of the initial mixture is taken out and replaced with the same quantity of water, then the quantity of milk in the resultant mixture will be 20 ml less than water.

Quantity II: Find the quantity of pure milk in the initial mixture. A vessel contains certain quantity of pure milk, if replaced with 20% water, then the difference between the quantity of milk and water in the resultant mixture will be 48 ml.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

12. Questions

Quantity I: Find the length of the train. The time taken by the train to cross the platform of length 200 meters with the speed of 40 m/sec is 7 seconds more than the time taken by a man to cross the same train running in opposite direction with the speed of 10 m/sec.

Quantity II: Find the sum of the length of bridge and tunnel. If the time taken by the train of length 300 meters to cross the bridge is 1 second more than the time taken by the same train to cross the tunnel. The speed of the train is 50 m/sec. The ratio of the length of bridge and tunnel is 3:2.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

13. Questions

Quantity I: Find the value of $[X/50]$. A person invested Rs. $[500X]$ in a bank at the rate of 20% per annum compounded annually and Rs. $[500X-10000]$ invested in another scheme at the rate of 18% per annum for simple interest. The total interest received from both the scheme together after 2 years is Rs. 36400.

Quantity II: Find the value of $[Y/2]$. A person invested Rs. 60000 at the rate of $[Y]\%$ per annum compounded annually for 2 years. If the same amount is invested at simple interest at same rate of interest for 3 years, then the simple interest received will be Rs. 5400 more than the compound interest. ($Y < 50$)

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

14. Questions

Quantity I: Find the time taken by 2 men and 3 women to complete $3/4^{\text{th}}$ of the total work, if the work done by 10 men in 20 days is equal to the work done by 40 women in 15 days.

Quantity II: Find the time taken by C to complete the whole work, if A and C together can complete the whole work in $(60/7)$ days, whereas B and C together takes $(45/4)$ days to complete the same work. The sum of the efficiency of A and B is 25.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II
- d. Quantity I $\leq$ Quantity II
- e. Quantity I = Quantity II

15. Questions

Quantity I: find the speed of the boat in still water. The ratio of the speed of boat in still water and the speed of the river is 4:1. The time taken by the boat to cover 300 km downstream is 1 hour less than the time taken by the same boat to cover 195 km upstream.

Quantity II: find the upstream speed of the boat. The speed of the boat in still water is 4 km/hr more than the upstream speed of the boat. The downstream distance covered by the boat in 5 hours is 14 km more than the upstream distance covered by the same boat in 6 hours.

- a. Quantity I > Quantity II
- b. Quantity I < Quantity II
- c. Quantity I $\geq$ Quantity II

d. Quantity I $\leq$ Quantity II

e. Quantity I = Quantity II

16. Questions

Read the following information carefully and answer the questions based on it.

Certain number of students (boys + girls) took admission in primary and secondary school in three different years 2011, 2012 and 2013.

2011: The total number of students who took admission is 1200, out of which $\frac{1}{4}^{\text{th}}$ took admission in secondary schools. The ratio of the number of girls who took admission in primary school and the number of boys who took admission in secondary school is 5:4. The number of girls who took admission in secondary school is 10% of the number of boys who took admission in primary school.

2012: The total number of students (boys + girls) who took admission in primary school is 33.33% more than the previous year. The total number of students (boys + girls) who took admission in secondary school is $\frac{2}{5}^{\text{th}}$ of the total number of students (primary + secondary) who took admission. The number of girls who took admission in secondary school is $\frac{2}{7}^{\text{th}}$ of the number of boys who took admission in primary school. The number of girls who took admission in primary school is 16.67% less than the number of boys who took admission in secondary school.

2013: The total number of students (primary + secondary) who took admission is 50% more than the previous year. The ratio of the total number of students (boys + girls) who took admission in primary and secondary school is 3:2. The number of girls who took admission in primary school is $\frac{3}{4}^{\text{th}}$ of the number of boys who took admission in secondary school. The number of girls who took admission in secondary school is 33.33% of the number of boys who took admission in primary school.

Note:

The total number of students who took admission = the number of students who admission primary school + the number of students who admission secondary school

The total number of students who took admission in primary school = the number of boys who took admission in primary school + the number of girls who took admission in primary school

The total number of students who took admission in secondary school = the number of boys who took admission in secondary school + the number of girls who took admission in secondary school

Find the average of the number of boys who took admission in secondary school in 2011 and the number of girls who took admission in primary school in 2013.

a. 420

b. 360

c. 500

d. 280

e. 120

17. Questions

In secondary school in 2012, out of the total number of girls who took admission, if 10% did not appear in the examination and the number of girls who failed in the examination is 30, then find the number of girls who passed in the examination in secondary school in 2012.

- a. 120
- b. 150
- c. 100
- d. 160
- e. 140

18. Questions

In another year 2014, the total number of students who took admission is __[?], while the number of students who took admission in secondary school is equals to 2012. The number of students who took admission in primary school in 2014 is 1600.

What should be the correct replacement for question mark?

- a. 5600
- b. 3200
- c. 2400
- d. 6000
- e. 7200

19. Questions

Find the sum of the number of boys who took admission in primary school in 2012 and the number of boys who took admission in secondary school in 2013.

- a. 2000
- b. 2500
- c. 3000
- d. 3500
- e. 1500

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20. Questions

In 2011, if $\frac{1}{6}$ th of the total number of girls who took admission in secondary school use own transport, which is 10% of the number of girls who took admission in primary school who use own transport, then find the difference between the total number of girls who use school transport in 2011. (consider school and own transport only)

- a. 240
- b. 150
- c. 300
- d. 120
- e. 280

Explanations:

1. Questions

The ratio of the cost price of articles P and Q = 1:2. Let it be Rs. a and Rs. 2a respectively.

Now,

The C.P. of article P = Rs. a

The S.P. of article P = 120% of a = Rs. 1.20a

The M.P. of article P = Rs. (1.20a + 30)

Also,

The C.P. of article Q = Rs. 2a

The S.P. of article Q = 140% of 2a = Rs. 2.80a

The M.P. of article Q = Rs. (2.80a + 40)

So,

$$2.80a - (1.20a + 30) = 130$$

$$2.80a - 130 = 1.20a + 30$$

$$1.60a = 160$$

$$a = 100$$

Now,

Let the cost price of article R is Rs. b

So, the cost price of article S = Rs. (b + 140)

Now,

The C.P. of article R = Rs. b

The S.P. of article R = 110% of b = Rs. 1.10b

The M.P. of article R = Rs. (1.10b + 54)

Also,

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The C.P. of article S = Rs. $(b + 140)$

The S.P. of article S = 125% of $(b + 140)$ = Rs. $(1.25b + 175)$

The M.P. of article S = $(1.25b + 175 + 68)$ = Rs. $(1.25b + 243)$

So,

$$(1.25b + 175) - (1.10b + 54) = 148$$

$$1.25b + 27 = 1.10b + 54$$

$$0.15b = 27$$

$$b = 180$$

Article P

The C.P. = a = Rs. 100

The S.P. = $1.20a$ = $(1.20 * 100)$ = Rs. 120

The M.P. = $(1.20a + 30)$ = $(120 + 30)$ = Rs. 150

Article Q

The C.P. = $2a$ = $(2 * 100)$ = Rs. 200

The S.P. = $2.80a$ = $(2.80 * 100)$ = Rs. 280

The M.P. = $(2.80a + 40)$ = $(280 + 40)$ = Rs. 320

Article R

The C.P. = b = Rs. 180

The S.P. = $1.10b$ = $(1.10 * 180)$ = Rs. 198

The M.P. = $(1.10b + 54)$ = $(198 + 54)$ = Rs. 252

Article S

The C.P. = $(b + 140)$ = $(180 + 140)$ = Rs. 320

The S.P. = $(1.25b + 175)$ = $(1.25 * 180 + 175)$ = $(225 + 175)$ = Rs. 400

The M.P. = $(1.25b + 243)$ = $(225 + 243)$ = Rs. 468

Article T

The ratio of the cost price and selling price = 8:9. (Note: 12.50% = 1/8)

Let the C.P. be Rs. 8c

So, the S.P. = Rs. 9c

And, the M.P. = $(9c + 50)$

Now,

$$8c/(9c + 50) = 100/125$$

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$$8c/(9c + 50) = 4/5$$

$$40c = 36c + 200$$

$$4c = 200$$

$$c = 50$$

Therefore,

$$\text{The C.P.} = 8c = (8 * 50) = \text{Rs. } 400$$

$$\text{The S.P.} = (9 * 50) = \text{Rs. } 450$$

$$\text{The M.P.} = (9c + 50) = (450 + 50) = \text{Rs. } 500$$

Article	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
P	100	150	120
Q	200	320	280
R	180	252	198
S	320	468	400
T	400	500	450

Answer: C

Now, according to the question,

Article U

$$\text{The C.P.} = 150\% \text{ of } 200 = \text{Rs. } 300$$

$$\text{The mark-up amount} = Y\% \text{ of } 300 = \text{Rs. } 3Y$$

$$\text{The M.P.} = \text{Rs. } (300 + 3Y)$$

$$\text{The S.P.} = 4/3 \text{ of } 300 = \text{Rs. } 400 \text{ (Note: } 33.33\% = 1/3)$$

Now,

$$(300 + 3Y) - 400 = 80$$

$$300 + 3Y = 480$$

$$3Y = 180$$

$$Y = 60$$

$$\text{Therefore, } (Y/12) = (60/12) = 5.$$

Hence, the correct answer is option C.

2. Questions

The ratio of the cost price of articles P and Q = 1:2. Let it be Rs. a and Rs. 2a respectively.

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Now,

The C.P. of article P = Rs. a

The S.P. of article P = 120% of a = Rs. 1.20a

The M.P. of article P = Rs. (1.20a + 30)

Also,

The C.P. of article Q = Rs. 2a

The S.P. of article Q = 140% of 2a = Rs. 2.80a

The M.P. of article Q = Rs. (2.80a + 40)

So,

$$2.80a - (1.20a + 30) = 130$$

$$2.80a - 130 = 1.20a + 30$$

$$1.60a = 160$$

$$a = 100$$

Now,

Let the cost price of article R is Rs. b

So, the cost price of article S = Rs. (b + 140)

Now,

The C.P. of article R = Rs. b

The S.P. of article R = 110% of b = Rs. 1.10b

The M.P. of article R = Rs. (1.10b + 54)

Also,

The C.P. of article S = Rs. (b + 140)

The S.P. of article S = 125% of (b + 140) = Rs. (1.25b + 175)

The M.P. of article S = (1.25b + 175 + 68) = Rs. (1.25b + 243)

So,

$$(1.25b + 175) - (1.10b + 54) = 148$$

$$1.25b + 27 = 1.10b + 54$$

$$0.15b = 27$$

$$b = 180$$

Article P

The C.P. = a = Rs. 100

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The S.P. = $1.20a = (1.20 * 100) = \text{Rs. } 120$

The M.P. = $(1.20a + 30) = (120 + 30) = \text{Rs. } 150$

Article Q

The C.P. = $2a = (2 * 100) = \text{Rs. } 200$

The S.P. = $2.80a = (2.80 * 100) = \text{Rs. } 280$

The M.P. = $(2.80a + 40) = (280 + 40) = \text{Rs. } 320$

Article R

The C.P. = $b = \text{Rs. } 180$

The S.P. = $1.10b = (1.10 * 180) = \text{Rs. } 198$

The M.P. = $(1.10b + 54) = (198 + 54) = \text{Rs. } 252$

Article S

The C.P. = $(b + 140) = (180 + 140) = \text{Rs. } 320$

The S.P. = $(1.25b + 175) = (1.25 * 180 + 175) = (225 + 175) = \text{Rs. } 400$

The M.P. = $(1.25b + 243) = (225 + 243) = \text{Rs. } 468$

Article T

The ratio of the cost price and selling price = 8:9. (Note: $12.50\% = 1/8$)

Let the C.P. be Rs. $8c$

So, the S.P. = Rs. $9c$

And, the M.P. = $(9c + 50)$

Now,

$$8c/(9c + 50) = 100/125$$

$$8c/(9c + 50) = 4/5$$

$$40c = 36c + 200$$

$$4c = 200$$

$$c = 50$$

Therefore,

The C.P. = $8c = (8 * 50) = \text{Rs. } 400$

The S.P. = $(9 * 50) = \text{Rs. } 450$

The M.P. = $(9c + 50) = (450 + 50) = \text{Rs. } 500$

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Article	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
P	100	150	120
Q	200	320	280
R	180	252	198
S	320	468	400
T	400	500	450

Answer: D

The C.P. of article Q = Rs. 200

The M.P. of article Q = Rs. 320

The mark-up amount of article Q = $(320 - 200) = \text{Rs. } 120$

So, the mark-up percentage of article Q = $(120/200) * 100 = 60\%$.

Now,

The C.P. of article S = Rs. 320

The M.P. of article S = Rs. 468

The mark-up amount of article S = $(468 - 320) = \text{Rs. } 148$

So, the mark-up percentage of article S = $(148/320) * 100 = 46.25\%$.

Therefore, the required difference = $(60 - 46.25) = 13.75$.

Hence, the correct answer is option D.

3. Questions

The ratio of the cost price of articles P and Q = 1:2. Let it be Rs. a and Rs. 2a respectively.

Now,

The C.P. of article P = Rs. a

The S.P. of article P = 120% of a = Rs. 1.20a

The M.P. of article P = Rs. $(1.20a + 30)$

Also,

The C.P. of article Q = Rs. 2a

The S.P. of article Q = 140% of 2a = Rs. 2.80a

The M.P. of article Q = Rs. $(2.80a + 40)$

So,

$$2.80a - (1.20a + 30) = 130$$

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$$2.80a - 130 = 1.20a + 30$$

$$1.60a = 160$$

$$a = 100$$

Now,

Let the cost price of article R is Rs. b

So, the cost price of article S = Rs. (b + 140)

Now,

The C.P. of article R = Rs. b

The S.P. of article R = 110% of b = Rs. 1.10b

The M.P. of article R = Rs. (1.10b + 54)

Also,

The C.P. of article S = Rs. (b + 140)

The S.P. of article S = 125% of (b + 140) = Rs. (1.25b + 175)

The M.P. of article S = (1.25b + 175 + 68) = Rs. (1.25b + 243)

So,

$$(1.25b + 175) - (1.10b + 54) = 148$$

$$1.25b + 27 = 1.10b + 54$$

$$0.15b = 27$$

$$b = 180$$

Article P

The C.P. = a = Rs. 100

The S.P. = 1.20a = (1.20 * 100) = Rs. 120

The M.P. = (1.20a + 30) = (120 + 30) = Rs. 150

Article Q

The C.P. = 2a = (2 * 100) = Rs. 200

The S.P. = 2.80a = (2.80 * 100) = Rs. 280

The M.P. = (2.80a + 40) = (280 + 40) = Rs. 320

Article R

The C.P. = b = Rs. 180

The S.P. = 1.10b = (1.10 * 180) = Rs. 198

The M.P. = (1.10b + 54) = (198 + 54) = Rs. 252

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Article S

The C.P. = $(b + 140) = (180 + 140) = \text{Rs. } 320$

The S.P. = $(1.25b + 175) = (1.25 * 180 + 175) = (225 + 175) = \text{Rs. } 400$

The M.P. = $(1.25b + 243) = (225 + 243) = \text{Rs. } 468$

Article T

The ratio of the cost price and selling price = 8:9. (**Note: 12.50% = 1/8**)

Let the C.P. be Rs. 8c

So, the S.P. = Rs. 9c

And, the M.P. = $(9c + 50)$

Now,

$$8c/(9c + 50) = 100/125$$

$$8c/(9c + 50) = 4/5$$

$$40c = 36c + 200$$

$$4c = 200$$

$$c = 50$$

Therefore,

The C.P. = $8c = (8 * 50) = \text{Rs. } 400$

The S.P. = $(9 * 50) = \text{Rs. } 450$

The M.P. = $(9c + 50) = (450 + 50) = \text{Rs. } 500$

Article	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
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Answer: A

Article R

The original C.P. = Rs. 180

The M.P. = Rs. 252

The mark-up amount = $(252 - 180) = \text{Rs. } 72$

So, the mark-up percentage = $(72/180) * 100 = 40\%$

Now,

The new C.P. = $(180 + 60) = \text{Rs. } 240$

The new M.P. = 140% of $240 = \text{Rs. } 336$

The new S.P. = $(336 - 36) = \text{Rs. } 300$

The original S.P. = $\text{Rs. } 198$

Therefore, the required ratio = $198:300 = 33:50$.

Hence, the correct answer is option A.

4. Questions

The ratio of the cost price of articles P and Q = 1:2. Let it be Rs. a and Rs. 2a respectively.

Now,

The C.P. of article P = Rs. a

The S.P. of article P = 120% of a = Rs. $1.20a$

The M.P. of article P = Rs. $(1.20a + 30)$

Also,

The C.P. of article Q = Rs. 2a

The S.P. of article Q = 140% of 2a = Rs. $2.80a$

The M.P. of article Q = Rs. $(2.80a + 40)$

So,

$$2.80a - (1.20a + 30) = 130$$

$$2.80a - 130 = 1.20a + 30$$

$$1.60a = 160$$

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Now,

Let the cost price of article R is Rs. b

So, the cost price of article S = Rs. $(b + 140)$

Now,

The C.P. of article R = Rs. b

The S.P. of article R = 110% of b = Rs. $1.10b$

The M.P. of article R = Rs. $(1.10b + 54)$

Also,

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The C.P. of article S = Rs. $(b + 140)$

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The M.P. of article S = $(1.25b + 175 + 68)$ = Rs. $(1.25b + 243)$

So,

$$(1.25b + 175) - (1.10b + 54) = 148$$

$$1.25b + 27 = 1.10b + 54$$

$$0.15b = 27$$

$$b = 180$$

Article P

The C.P. = a = Rs. 100

The S.P. = $1.20a$ = $(1.20 * 100)$ = Rs. 120

The M.P. = $(1.20a + 30)$ = $(120 + 30)$ = Rs. 150

Article Q

The C.P. = $2a$ = $(2 * 100)$ = Rs. 200

The S.P. = $2.80a$ = $(2.80 * 100)$ = Rs. 280

The M.P. = $(2.80a + 40)$ = $(280 + 40)$ = Rs. 320

Article R

The C.P. = b = Rs. 180

The S.P. = $1.10b$ = $(1.10 * 180)$ = Rs. 198

The M.P. = $(1.10b + 54)$ = $(198 + 54)$ = Rs. 252

Article S

The C.P. = $(b + 140)$ = $(180 + 140)$ = Rs. 320

The S.P. = $(1.25b + 175)$ = $(1.25 * 180 + 175)$ = $(225 + 175)$ = Rs. 400

The M.P. = $(1.25b + 243)$ = $(225 + 243)$ = Rs. 468

Article T

The ratio of the cost price and selling price = 8:9. (Note: 12.50% = 1/8)

Let the C.P. be Rs. 8c

So, the S.P. = Rs. 9c

And, the M.P. = $(9c + 50)$

Now,

$$8c / (9c + 50) = 100 / 125$$

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$$8c/(9c + 50) = 4/5$$

$$40c = 36c + 200$$

$$4c = 200$$

$$c = 50$$

Therefore,

$$\text{The C.P.} = 8c = (8 * 50) = \text{Rs. } 400$$

$$\text{The S.P.} = (9 * 50) = \text{Rs. } 450$$

$$\text{The M.P.} = (9c + 50) = (450 + 50) = \text{Rs. } 500$$

Article	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
P	100	150	120
Q	200	320	280
R	180	252	198
S	320	468	400
T	400	500	450

Answer: E

The S.P. of article P = Rs. 120

The M.P. of article T = Rs. 500

Therefore, the required percentage = $(120/500) * 100 = 24\%$.

Hence, the correct answer is option E.

5. Questions

The ratio of the cost price of articles P and Q = 1:2. Let it be Rs. a and Rs. 2a respectively.

Now,

The C.P. of article P = Rs. a

The S.P. of article P = 120% of a = Rs. 1.20a

The M.P. of article P = Rs. (1.20a + 30)

Also,

The C.P. of article Q = Rs. 2a

The S.P. of article Q = 140% of 2a = Rs. 2.80a

The M.P. of article Q = Rs. (2.80a + 40)

So,

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$$2.80a - (1.20a + 30) = 130$$

$$2.80a - 130 = 1.20a + 30$$

$$1.60a = 160$$

$$a = 100$$

Now,

Let the cost price of article R is Rs. b

So, the cost price of article S = Rs. (b + 140)

Now,

The C.P. of article R = Rs. b

The S.P. of article R = 110% of b = Rs. 1.10b

The M.P. of article R = Rs. (1.10b + 54)

Also,

The C.P. of article S = Rs. (b + 140)

The S.P. of article S = 125% of (b + 140) = Rs. (1.25b + 175)

The M.P. of article S = (1.25b + 175 + 68) = Rs. (1.25b + 243)

So,

$$(1.25b + 175) - (1.10b + 54) = 148$$

$$1.25b + 27 = 1.10b + 54$$

$$0.15b = 27$$

$$b = 180$$

Article P

The C.P. = a = Rs. 100

The S.P. = 1.20a = (1.20 * 100) = Rs. 120

The M.P. = (1.20a + 30) = (120 + 30) = Rs. 150

Article Q

The C.P. = 2a = (2 * 100) = Rs. 200

The S.P. = 2.80a = (2.80 * 100) = Rs. 280

The M.P. = (2.80a + 40) = (280 + 40) = Rs. 320

Article R

The C.P. = b = Rs. 180

The S.P. = 1.10b = (1.10 * 180) = Rs. 198

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The M.P. = $(1.10b + 54) = (198 + 54) = \text{Rs. } 252$

Article S

The C.P. = $(b + 140) = (180 + 140) = \text{Rs. } 320$

The S.P. = $(1.25b + 175) = (1.25 * 180 + 175) = (225 + 175) = \text{Rs. } 400$

The M.P. = $(1.25b + 243) = (225 + 243) = \text{Rs. } 468$

Article T

The ratio of the cost price and selling price = 8:9. (**Note: 12.50% = 1/8**)

Let the C.P. be Rs. 8c

So, the S.P. = Rs. 9c

And, the M.P. = $(9c + 50)$

Now,

$$8c/(9c + 50) = 100/125$$

$$8c/(9c + 50) = 4/5$$

$$40c = 36c + 200$$

$$4c = 200$$

$$c = 50$$

Therefore,

The C.P. = $8c = (8 * 50) = \text{Rs. } 400$

The S.P. = $(9 * 50) = \text{Rs. } 450$

The M.P. = $(9c + 50) = (450 + 50) = \text{Rs. } 500$

Article	The cost price (in Rs.)	The marked price (in Rs.)	The selling price (in Rs.)
P	100	150	120
Q	200	320	280
R	180	252	198
S	320	468	400
T	400	500	450

Answer: C

Article P

The C.P. = Rs. 100

The M.P. = Rs. 150

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The S.P. after first discount = Rs. 120

The additional discount = $D\%$ of 120 = Rs. $1.20D$

So, the S.P. after additional discount = Rs. $(120 - 1.20D)$

Now,

$$(120 - 1.20D) - 100 = (D - 2)$$

$$20 - 1.20D = D - 2$$

$$2.20D = 22$$

$$D = 10$$

I: $M = 8$ and $N = 12$

Here, 10 lies between 8 and 12.

So, I does satisfy.

II: $M = 5$ and $N = 9$

Here, 10 does not lies between 5 and 9.

So, II does not satisfy.

III: $M = 6$ and $N = 15$

Here, 10 lies between 6 and 15.

So, III does satisfy.

Therefore, both I and III follow.

Hence, the correct answer is option C.

6. Questions

Let the total number of defective items manufactured by all four shops together be 100k.

Now,

The total number of items manufactured by shop A = 120

The number of defective items manufactured by shop A = 24% of 100k = 24k

So, the number of non-defective items manufactured by shop A = $(120 - 24k)$

The number of non-defective items sold by shop A = 72

So, the number of non-defective items unsold by shop A = $(120 - 24k) - 72 = (48 - 24k)$

Also,

The total number of items manufactured by shop B = 112

The number of defective items manufactured by shop B = 28% of 100k = 28k

So, the number of non-defective items manufactured by shop B = $(112 - 28k)$

The number of non-defective items unsold by shop B = $(72 - 44) = 28$

So, the number of non-defective items sold by shop B = $(112 - 28k) - 28 = (84 - 28k)$

Now,

$$(48 - 24k) / (84 - 28k) = 3/7$$

$$336 - 168k = 252 - 84k$$

$$84k = 84$$

$$k = 1$$

Therefore, the total number of defective items manufactured by all five shops together = $100k = (100 * 1) = 100$.

Shop A

The total number of items manufactured = 120

The number of defective items manufactured = 24% of 100 = 24

The number of non-defective items manufactured = $(120 - 24) = 96$

The number of non-defective items sold = 72

The number of non-defective items unsold = $(96 - 72) = 24$

Shop B

The total number of items manufactured = 112

The number of defective items manufactured = 28% of 100 = 28

The number of non-defective items manufactured = $(112 - 28) = 84$

The number of non-defective items unsold = 28

The number of non-defective items sold = $(84 - 28) = 56$

Shop C

The total number of items manufactured = 108

The number of defective items manufactured = 18% of 100 = 18

The number of non-defective items manufactured = $(108 - 18) = 90$

The number of non-defective items sold = $(90 - 40) = 50$

The number of non-defective items unsold = $(90 - 50) = 40$

Shop D

The total number of items manufactured = 80

The number of defective items manufactured = 20% of 100 = 20

The number of non-defective items manufactured = $(80 - 20) = 60$

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The number of non-defective items unsold = 24

The number of non-defective items sold = $(60 - 24) = 36$

Shop E

The total number of items manufactured = 50

The number of defective items manufactured = 10% of 100 = 10

The number of non-defective items manufactured = $(50 - 10) = 40$

The number of non-defective items sold = $(40 + 24) / 2 = 64/2 = 32$

The number of non-defective items unsold = $(40 - 32) = 8$

Shop	The total number of items manufactured	The number of defective items manufactured	The number of non-defective items manufactured	The number of non-defective items sold	The number of non-defective items unsold
A	120	24	96	72	24
B	112	28	84	56	28
C	108	18	90	50	40
D	80	20	60	36	24
E	50	10	40	32	8
Total	470	100	370	246	124

Answer: E

Now, according to the question,

The total number of non-defective items sold by shop D = 36

The total number of non-defective items sold by shop E = 32

Let the price of each item in shop D and E be Rs. $4n$ and Rs. $3n$ respectively.

So, the total revenue generated from the sale of all non-defective items by shop D = $(36 * 4n) = \text{Rs. } 144n$

And, the total revenue generated from the sale of all non-defective items by shop E = $(32 * 3n) = \text{Rs. } 96n$

Now,

$$144n - 96n = 4800$$

$$48n = 4800$$

$$n = 100$$

Therefore, the difference between the price of each item in shop D and E = $(4n - 3n) = n = \text{Rs. } 100$.

Hence, the correct answer is option E.

7. Questions

Let the total number of defective items manufactured by all four shops together be 100k.

Now,

The total number of items manufactured by shop A = 120

The number of defective items manufactured by shop A = 24% of 100k = 24k

So, the number of non-defective items manufactured by shop A = $(120 - 24k)$

The number of non-defective items sold by shop A = 72

So, the number of non-defective items unsold by shop A = $(120 - 24k) - 72 = (48 - 24k)$

Also,

The total number of items manufactured by shop B = 112

The number of defective items manufactured by shop B = 28% of 100k = 28k

So, the number of non-defective items manufactured by shop B = $(112 - 28k)$

The number of non-defective items unsold by shop B = $(72 - 44) = 28$

So, the number of non-defective items sold by shop B = $(112 - 28k) - 28 = (84 - 28k)$

Now,

$$(48 - 24k) / (84 - 28k) = 3/7$$

$$336 - 168k = 252 - 84k$$

$$84k = 84$$

$$k = 1$$

Therefore, the total number of defective items manufactured by all five shops together = $100k = (100 * 1) = 100$.

Shop A

The total number of items manufactured = 120

The number of defective items manufactured = 24% of 100 = 24

The number of non-defective items manufactured = $(120 - 24) = 96$

The number of non-defective items sold = 72

The number of non-defective items unsold = $(96 - 72) = 24$

Shop B

The total number of items manufactured = 112

The number of defective items manufactured = 28% of 100 = 28

The number of non-defective items manufactured = $(112 - 28) = 84$

The number of non-defective items unsold = 28

The number of non-defective items sold = $(82 - 28) = 56$

Shop C

The total number of items manufactured = 108

The number of defective items manufactured = 18% of 100 = 18

The number of non-defective items manufactured = $(108 - 18) = 90$

The number of non-defective items sold = $(56 - 6) = 50$

The number of non-defective items unsold = $(90 - 50) = 40$

Shop D

The total number of items manufactured = 80

The number of defective items manufactured = 20% of 100 = 20

The number of non-defective items manufactured = $(80 - 20) = 60$

The number of non-defective items unsold = 24

The number of non-defective items sold = $(60 - 24) = 36$

Shop E

The total number of items manufactured = 50

The number of defective items manufactured = 10% of 100 = 10

The number of non-defective items manufactured = $(50 - 10) = 40$

The number of non-defective items sold = $(40 + 24) / 2 = 64/2 = 32$

The number of non-defective items unsold = $(40 - 32) = 8$

Shop	The total number of items manufactured	The number of defective items manufactured	The number of non-defective items manufactured	The number of non-defective items sold	The number of non-defective items unsold
A	120	24	96	72	24
B	112	28	84	56	28
C	108	18	90	50	40
D	80	20	60	36	24
E	50	10	40	32	8
Total	470	100	370	246	124

Answer: C

Shop A

The total number of non-defective items manufactured = 96

The number of non-defective items unsold = 24

So, the required percentage = $(24/96) \times 100 = 25\%$

Shop B

The total number of non-defective items manufactured = 84

The number of non-defective items unsold = 28

So, the required percentage = $(28/84) \times 100 = 33.33\%$

Shop C

The total number of non-defective items manufactured = 90

The number of non-defective items unsold = 40

So, the required percentage = $(40/90) \times 100 = 44.44\%$

Shop D

The total number of non-defective items manufactured = 60

The number of non-defective items unsold = 24

So, the required percentage = $(24/60) \times 100 = 40\%$

Shop E

The total number of non-defective items manufactured = 40

The number of non-defective items unsold = 8

So, the required percentage = $(8/40) \times 100 = 20\%$

Therefore, shop C shows the highest percentage = 44.44%

Hence, the correct answer is option C.

8. Questions

Let the total number of defective items manufactured by all four shops together be 100k.

Now,

The total number of items manufactured by shop A = 120

The number of defective items manufactured by shop A = 24% of 100k = 24k

So, the number of non-defective items manufactured by shop A = $(120 - 24k)$

The number of non-defective items sold by shop A = 72

So, the number of non-defective items unsold by shop A = $(120 - 24k) - 72 = (48 - 24k)$

Also,

The total number of items manufactured by shop B = 112

The number of defective items manufactured by shop B = 28% of 100k = 28k

So, the number of non-defective items manufactured by shop B = (112 – 28k)

The number of non-defective items unsold by shop B = (72 – 44) = 28

So, the number of non-defective items sold by shop B = (112 – 28k) – 28 = (84 – 28k)

Now,

$$(48 - 24k) / (84 - 28k) = 3/7$$

$$336 - 168k = 252 - 84k$$

$$84k = 84$$

$$k = 1$$

Therefore, the total number of defective items manufactured by all five shops together = 100k = (100 * 1) = 100.

Shop A

The total number of items manufactured = 120

The number of defective items manufactured = 24% of 100 = 24

The number of non-defective items manufactured = (120 – 24) = 96

The number of non-defective items sold = 72

The number of non-defective items unsold = (96 – 72) = 24

Shop B

The total number of items manufactured = 112

The number of defective items manufactured = 28% of 100 = 28

The number of non-defective items manufactured = (112 – 28) = 84

The number of non-defective items unsold = 28

The number of non-defective items sold = (82 – 28) = 56

Shop C

The total number of items manufactured = 108

The number of defective items manufactured = 18% of 100 = 18

The number of non-defective items manufactured = (108 – 18) = 90

The number of non-defective items sold = (56-6) = 50

The number of non-defective items unsold = (90 – 50) = 40

Shop D

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The total number of items manufactured = 80

The number of defective items manufactured = 20% of 100 = 20

The number of non-defective items manufactured = $(80 - 20) = 60$

The number of non-defective items unsold = 24

The number of non-defective items sold = $(60 - 24) = 36$

Shop E

The total number of items manufactured = 50

The number of defective items manufactured = 10% of 100 = 10

The number of non-defective items manufactured = $(50 - 10) = 40$

The number of non-defective items sold = $(40 + 24) / 2 = 64/2 = 32$

The number of non-defective items unsold = $(40 - 32) = 8$

Shop	The total number of items manufactured	The number of defective items manufactured	The number of non-defective items manufactured	The number of non-defective items sold	The number of non-defective items unsold
A	120	24	96	72	24
B	112	28	84	56	28
C	108	18	90	50	40
D	80	20	60	36	24
E	50	10	40	32	8
Total	470	100	370	246	124

Answer: B

The number of non-defective items sold by shop C = 50

The number of non-defective items unsold by shop A = 24

Therefore, the required ratio = $50:24 = 25:12$.

Hence, the correct answer is option B.

9. Questions

Let the total number of defective items manufactured by all four shops together be 100k.

Now,

The total number of items manufactured by shop A = 120

The number of defective items manufactured by shop A = 24% of 100k = 24k

@QuizByMickey

So, the number of non-defective items manufactured by shop A = $(120 - 24k)$

The number of non-defective items sold by shop A = 72

So, the number of non-defective items unsold by shop A = $(120 - 24k) - 72 = (48 - 24k)$

Also,

The total number of items manufactured by shop B = 112

The number of defective items manufactured by shop B = 28% of 100k = 28k

So, the number of non-defective items manufactured by shop B = $(112 - 28k)$

The number of non-defective items unsold by shop B = $(72 - 44) = 28$

So, the number of non-defective items sold by shop B = $(112 - 28k) - 28 = (84 - 28k)$

Now,

$$(48 - 24k) / (84 - 28k) = 3/7$$

$$336 - 168k = 252 - 84k$$

$$84k = 84$$

$$k = 1$$

Therefore, the total number of defective items manufactured by all five shops together = $100k = (100 * 1) = 100$.

Shop A

The total number of items manufactured = 120

The number of defective items manufactured = 24% of 100 = 24

The number of non-defective items manufactured = $(120 - 24) = 96$

The number of non-defective items sold = 72

The number of non-defective items unsold = $(96 - 72) = 24$

Shop B

The total number of items manufactured = 112

The number of defective items manufactured = 28% of 100 = 28 @QuizByMickey

The number of non-defective items manufactured = $(112 - 28) = 84$

The number of non-defective items unsold = 28

The number of non-defective items sold = $(84 - 28) = 56$

Shop C

The total number of items manufactured = 108

The number of defective items manufactured = 18% of 100 = 18

The number of non-defective items manufactured = $(108 - 18) = 90$

The number of non-defective items sold = $(56 - 6) = 50$

The number of non-defective items unsold = $(90 - 50) = 40$

Shop D

The total number of items manufactured = 80

The number of defective items manufactured = 20% of 100 = 20

The number of non-defective items manufactured = $(80 - 20) = 60$

The number of non-defective items unsold = 24

The number of non-defective items sold = $(60 - 24) = 36$

Shop E

The total number of items manufactured = 50

The number of defective items manufactured = 10% of 100 = 10

The number of non-defective items manufactured = $(50 - 10) = 40$

The number of non-defective items sold = $(40 + 24) / 2 = 64 / 2 = 32$

The number of non-defective items unsold = $(40 - 32) = 8$

Shop	The total number of items manufactured	The number of defective items manufactured	The number of non-defective items manufactured	The number of non-defective items sold	The number of non-defective items unsold
A	120	24	96	72	24
B	112	28	84	56	28
C	108	18	90	50	40
D	80	20	60	36	24
E	50	10	40	32	8
Total	470	100	370	246	124

Answer: D

Shop A

The total number of defective items manufactured = 24

The total number of non-defective items manufactured = 96

The number of defective wooden items manufactured = 6

So, the number of non-defective wooden items manufactured = $(6 * 6) = 36$

Also, the number of defective steel items manufactured = $(24 - 6) = 18$

And, the number of non-defective steel items manufactured = $(96 - 36) = 60$

Therefore, the required ratio = $18:60 = 3:10$.

Hence, the correct answer is option D.

10. Questions

Let the total number of defective items manufactured by all four shops together be 100k.

Now,

The total number of items manufactured by shop A = 120

The number of defective items manufactured by shop A = 24% of 100k = 24k

So, the number of non-defective items manufactured by shop A = $(120 - 24k)$

The number of non-defective items sold by shop A = 72

So, the number of non-defective items unsold by shop A = $(120 - 24k) - 72 = (48 - 24k)$

Also,

The total number of items manufactured by shop B = 112

The number of defective items manufactured by shop B = 28% of 100k = 28k

So, the number of non-defective items manufactured by shop B = $(112 - 28k)$

The number of non-defective items unsold by shop B = $(72 - 44) = 28$

So, the number of non-defective items sold by shop B = $(112 - 28k) - 28 = (84 - 28k)$

Now,

$$(48 - 24k) / (84 - 28k) = 3/7$$

$$336 - 168k = 252 - 84k$$

$$84k = 84$$

$$k = 1$$

Therefore, the total number of defective items manufactured by all five shops together = $100k = (100 * 1) = 100$.

Shop A

The total number of items manufactured = 120

The number of defective items manufactured = 24% of 100 = 24

The number of non-defective items manufactured = $(120 - 24) = 96$

The number of non-defective items sold = 72

The number of non-defective items unsold = $(96 - 72) = 24$

Shop B

The total number of items manufactured = 112

The number of defective items manufactured = 28% of 100 = 28

The number of non-defective items manufactured = $(112 - 28) = 84$

The number of non-defective items unsold = 28

The number of non-defective items sold = $(82 - 28) = 56$

Shop C

The total number of items manufactured = 108

The number of defective items manufactured = 18% of 100 = 18

The number of non-defective items manufactured = $(108 - 18) = 90$

The number of non-defective items sold = $(56 - 6) = 50$

The number of non-defective items unsold = $(90 - 50) = 40$

Shop D

The total number of items manufactured = 80

The number of defective items manufactured = 20% of 100 = 20

The number of non-defective items manufactured = $(80 - 20) = 60$

The number of non-defective items unsold = 24

The number of non-defective items sold = $(60 - 24) = 36$

Shop E

The total number of items manufactured = 50

The number of defective items manufactured = 10% of 100 = 10

The number of non-defective items manufactured = $(50 - 10) = 40$

The number of non-defective items sold = $(40 + 24) / 2 = 64 / 2 = 32$

The number of non-defective items unsold = $(40 - 32) = 8$

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Shop	The total number of items manufactured	The number of defective items manufactured	The number of non-defective items manufactured	The number of non-defective items sold	The number of non-defective items unsold
A	120	24	96	72	24
B	112	28	84	56	28
C	108	18	90	50	40
D	80	20	60	36	24
E	50	10	40	32	8
Total	470	100	370	246	124

Answer: A

The total number of non-defective items sold by shop C = 50

The number of non-defective items sold to females by shop C = 40% of 50 = 20

And, the number of non-defective items sold to males by shop C = (50 – 20) = 30

So, the number of non-defective items sold to females by shop B = 20

And, the total number of non-defective items sold by shop B = 56

So, the number of non-defective items sold to males by shop B = (56 – 20) = 36

Therefore, the total number of non-defective items sold to males by shop B and C together = (36+30) = 66.

Hence, the correct answer is option A.

11. Questions

Answer: E

Quantity I

Let the initial quantity of pure milk and water be 3m ml and 2m ml respectively.

The total initial mixture = (3m + 2m) = 5m ml

Now, the quantity of milk in the resultant mixture = 75% of 3m = 2.25m

And, the quantity of water in the resultant mixture = (75% of 2m) + (25% of 5m) = (1.50m + 1.25m) = 2.75m ml

So,

$$2.75m - 2.25m = 20$$

$$0.50m = 20$$

$$m = 40$$

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Therefore, the initial quantity of water in the vessel = $2m = (2 * 40) = 80$ ml

Quantity I = 80

Quantity II

Let the initial quantity of pure milk be $100n$ ml

So, the quantity of water in the resultant mixture = 20% of $100n = 20n$ ml

And, the quantity of milk in the resultant mixture = 80% of $100n = 80n$ ml

Now,

$$80n - 20n = 48$$

$$60n = 48$$

$$n = 0.80$$

So, the quantity of pure milk in the initial mixture = $100n = (100 * 0.80) = 80$ ml

Therefore, Quantity II = 80

So, Quantity I = Quantity II.

Hence, the correct answer is option E.

12. Questions

Answer: A

Quantity I

Let the length of the train be L meters

The sum of the length of the train and the platform = $(L + 200)$ meters

And, the sum of the speed of the train and the man = $(40+10) = 50$ m/sec

Now,

$$(L + 200)/40 - (L/50) = 7$$

$$(5L + 1000) - 4L/200 = 7$$

$$L + 1000 = 1400$$

$$L = 400$$

So, the length of the train = 400 meters

Quantity I = 400.

Quantity II

Let the length of bridge and tunnel be $3T$ meters and $2T$ meters respectively.

So, the sum of the length of the train and bridge = $(300 + 3T)$ meters

And, the sum of the length of the train and tunnel = $(300 + 2T)$ meters

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Now,

$$(300 + 3T)/50 - (300 + 2T)/50 = 1$$

$$T/50 = 1$$

$$T = 50$$

So, the sum of the length of bridge and tunnel = $(3T + 2T) = 5T = (5 * 50) = 250$ meters.

Quantity II = 250.

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

13. Questions

Answer: B

Quantity I

Compound interest

The amount invested = Rs. 500X

CI rate = 20% p.a., compounded annually, N = 2 years

$$\text{So, } (20 + 20) + (20 * 20)/100 = (40 + 4)\% = 44\%$$

So, the compound interest received = 44% of 500X = Rs. 220X

simple interest

The amount invested = Rs. (500X – 10000)

SI rate = 18% p.a.

N = 2 years

So, the simple interest received = 36% of (500X – 10000) = Rs. (180X – 3600)

Now,

$$220X + (180X - 3600) = 36400$$

$$400X = 40000$$

$$X = 100$$

$$\text{Quantity I} = (X/50) = (100/50) = 2.$$

Quantity II

Compound interest

The amount invested = Rs. 60000

CI rate = Y% p.a., compounded annually, N = 2 years

$$\text{So, } (Y + Y) + (Y * Y)/100 = (2Y + Y^2/100)\%$$

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the compound interest = $(2Y + Y^2/100)\%$ of 60000 = Rs. $(1200Y + 6Y^2)$

simple interest

SI rate = $Y\%$ p.a.

$N = 3$ years

So, the simple interest = $3Y\%$ of 60000 = Rs. $1800Y$

Now,

$$1800Y - (1200Y + 6Y^2) = 5400$$

$$1800Y - 5400 = 1200Y + 6Y^2$$

$$6Y^2 - 600Y + 5400 = 0$$

$$Y^2 - 100Y + 900 = 0$$

$$Y^2 - 90Y - 10Y + 900 = 0$$

$$Y(Y - 90) - 10(Y - 90) = 0$$

$$(Y - 10)(Y - 90) = 0$$

$$Y = 10 \text{ or } 90$$

So, $Y = 10 (< 50)$

$$\text{Quantity II} = (Y/2) = (10/2) = 5.$$

Therefore, Quantity I < Quantity II.

Hence, the correct answer is option B.

14. Questions

Answer: A

Quantity I

Let the efficiency of each man and each woman be M and W respectively.

Now,

$$(M * 10 * 20) = (W * 40 * 15)$$

$$200M = 600W$$

$$M:W = 3:1$$

$$\text{So, the total work} = (M * 10 * 20) = (3 * 10 * 20) = 600$$

$$\text{Now, the combined efficiency of 2 men and 3 women} = (2 * 3) + (3 * 1) = (6 + 3) = 9$$

$$\text{Quantity I} = (3/4 \text{ of } 600)/9 = (450/9) = 50$$

Quantity II

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The time taken to complete the whole work by,

$$(A + C) = (60/7) \text{ days}$$

$$(B + C) = (45/4) \text{ days}$$

The total work = 180 units
 (L.C.M. of the numerator
 of the time taken)

So, the efficiency of,

$$(A + C) = 180 / (60/7) = 21$$

$$(B + C) = 180 / (45/4) = 16$$

$$A - B = (21 - 16) = 5 \dots\dots\dots(i)$$

$$A + B = 25 \dots\dots\dots(ii)$$

By solving equation (i) and (ii), we get,

$$2B = 20$$

$$B = 10$$

Now, put the value of B in equation (i), we get,

$$A - B = 5$$

$$A - 10 = 5$$

$$A = 15$$

So, the individual efficiency of C = $(21 - 15) = 6$

Now, the time taken by C to complete the whole work alone = $(180/6) = 30$ days.

Quantity II = 30

Therefore, Quantity I > Quantity II.

Hence, the correct answer is option A.

15. Questions

Answer: B

Quantity I

Let the speed of the boat in still water be $4v$ km/hr

So, the speed of the river = v km/hr

The downstream speed = $(4v + v) = 5v$ km/hr

The upstream speed = $(4v - v) = 3v$ km/hr

Now,

$$(195/3v) - (300/5v) = 1$$

$$(975 - 900)/15v = 1$$

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$$15v = 75$$

$$v = 5$$

So, the speed of the boat in still water = $4v = (4 * 5) = 20$ km

Quantity I = 20.

Quantity II

Let the upstream speed be x km/hr

So, the speed of the boat in still water = $(x + 4)$ km/hr

Let the speed of the river be y km/hr

Now,

$$(x + 4) - y = x$$

$$x + y = x + 4$$

$$y = 4$$

So, the speed of the river = $y = 4$ km/hr

The downstream speed = $(x + 4) + y = (x + 4) + 4 = (x + 8)$ km/hr

Now,

$$5 * (x + 8) - (6 * x) = 14$$

$$(5x + 40) - 6x = 14$$

$$40 - x = 14$$

$$x = 26$$

So, the upstream speed of the boat = $x = 26$ km/hr

Quantity II = 26.

Therefore, Quantity I < Quantity II.

Hence, the correct answer is option B.

16. Questions

2011

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The total number of students who took admission = 1200

The total number of students who took admission in secondary school = $1/4$ of 1200 = 300

The total number of students who took admission in primary school = $(1200 - 300) = 900$

Let the number of girls who took admission in primary school and boys who took admission in secondary school be $5a$ and $4a$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = $10:100 = 1:10$. Let it be b and $10b$ respectively.

Now,

$$5a + 10b = 900$$

$$a + 2b = 180 \dots\dots\dots(I)$$

And,

$$4a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$20a + 40b = 3600$$

$$20a + 5b = 1500$$

$$35b = 2100$$

$$b = 60$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 180$$

$$a + (2 * 60) = 180$$

$$a = 180 - 120$$

$$a = 60$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 10b = (10 * 60) = 600$$

$$\text{The number of girls who took admission in primary school} = 5a = (5 * 60) = 300$$

$$\text{The number of boys who took admission in secondary school} = 4a = (4 * 60) = 240$$

$$\text{The number of girls who took admission in secondary school} = b = 60$$

2012

$$\text{The total number of students who took admission in primary school} = 4/3 \text{ of } 900 = 1200 \text{ (Note: } 33.33\% = 1/3)$$

$$\text{The ratio of the total number of students who took admission in primary and secondary school} = 3:2$$

$$\text{The total number of students who took admission in secondary school} = 1200/3 * 2 = 800$$

$$\text{The total number of students who took admission} = (1200 + 800) = 2000$$

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 2:7. Let it be 2m and 7m respectively.

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 5:6. Let it be 5n and 6n respectively. (Note: 16.67% = 1/6)

Now,

$$7m + 5n = 1200 \dots\dots\dots(III)$$

And,

$$2m + 6n = 800$$

$$m + 3n = 400 \dots \dots \dots (IV)$$

By solving equations (III) and (IV), we get,

$$14m + 10n = 2400$$

$$7m + 5n = 1200$$

$$7m + 21n = 2800$$

$$16n = 1600$$

$$n = 100$$

Now, put the value of n in equation (III), we get,

$$7m + 5n = 1200$$

$$7m + (5 * 100) = 1200$$

$$7m = 1200 - 500$$

$$7m = 700$$

$$m = 100$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 7m = (7 * 100) = 700$$

$$\text{The number of girls who took admission in primary school} = 5n = (5 * 100) = 500$$

$$\text{The number of boys who took admission in secondary school} = 6n = (6 * 100) = 600$$

$$\text{The number of girls who took admission in secondary school} = 2m = (2 * 100) = 200$$

2013

$$\text{The total number of students who took admission} = 150\% \text{ of } 2000 = 3000$$

$$\text{The total number of students who took admission in primary school} = 3000/5*3 = 1800$$

$$\text{The total number of students who took admission in secondary school} = 3000/5*2 = 1200$$

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 3:4. Let it be 3p and 4p respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 1:3. Let it be q and 3q respectively.

Now,

$$3p + 3q = 1800$$

$$p + q = 600 \dots \dots \dots (V)$$

Also,

$$4p + q = 1200 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$12p + 12q = 7200$$

$$12p + 3q = 3600$$

$$9q = 3600$$

$$q = 400$$

Now, put the value of q in equation (V), we get,

$$p + q = 600$$

$$p + 400 = 600$$

$$p = 200$$

Therefore,

The number of boys who took admission in primary school = $3q = (3 * 400) = 1200$

The number of girls who took admission in primary school = $3p = (3 * 200) = 600$

The number of boys who took admission in secondary school = $4p = (4 * 200) = 800$

The number of girls who took admission in secondary school = $q = 400$

Year	The total number of students who took admission	The total number of students who took admission in primary school	The total number of students who took admission in secondary school	The number of boys who took admission in primary school	The number of girls who took admission in primary school	The number of boys who took admission in secondary school	The number of girls who took admission in secondary school
2011	1200	900	300	600	300	240	60
2012	2000	1200	800	700	500	600	200
2013	3000	1800	1200	1200	600	800	400
Total	6200	3900	2300	2500	1400	1640	660

Answer: A

Now, according to the question,

The number of boys who took admission in secondary school in 2011 = 240

The number of girls who took admission in primary school in 2013 = 600

Therefore, the required average = $(240 + 600)/2 = 840/2 = 420$.

Hence, the correct answer is option A.

17. Questions

2011

The total number of students who took admission = 1200

The total number of students who took admission in secondary school = $1/4$ of 1200 = 300

The total number of students who took admission in primary school = $(1200 - 300) = 900$

Let the number of girls who took admission in primary school and boys who took admission in secondary school be $5a$ and $4a$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = $10:100 = 1:10$. Let it be b and $10b$ respectively.

Now,

$$5a + 10b = 900$$

$$a + 2b = 180 \dots\dots\dots(I)$$

And,

$$4a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$20a + 40b = 3600$$

$$20a + 5b = 1500$$

$$35b = 2100$$

$$b = 60$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 180$$

$$a + (2 * 60) = 180$$

$$a = 180 - 120$$

$$a = 60$$

Therefore,

The number of boys who took admission in primary school = $10b = (10 * 60) = 600$

The number of girls who took admission in primary school = $5a = (5 * 60) = 300$

The number of boys who took admission in secondary school = $4a = (4 * 60) = 240$

The number of girls who took admission in secondary school = $b = 60$

2012

The total number of students who took admission in primary school = $4/3$ of 900 = 1200 (Note: 33.33% =

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1/3)

The ratio of the total number of students who took admission in primary and secondary school = 3:2

The total number of students who took admission in secondary school = $1200/3 \times 2 = 800$

The total number of students who took admission = $(1200 + 800) = 2000$

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 2:7. Let it be $2m$ and $7m$ respectively.

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 5:6. Let it be $5n$ and $6n$ respectively. **(Note: $16.67\% = 1/6$)**

Now,

$$7m + 5n = 1200 \dots \dots \dots (III)$$

And,

$$2m + 6n = 800$$

$$m + 3n = 400 \dots \dots \dots (IV)$$

By solving equations (III) and (IV), we get,

$$14m + 10n = 2400$$

$$7m + 5n = 1200$$

$$7m + 21n = 2800$$

$$16n = 1600$$

$$n = 100$$

Now, put the value of n in equation (III), we get,

$$7m + 5n = 1200$$

$$7m + (5 \times 100) = 1200$$

$$7m = 1200 - 500$$

$$7m = 700$$

$$m = 100$$

Therefore,

The number of boys who took admission in primary school = $7m = (7 \times 100) = 700$

The number of girls who took admission in primary school = $5n = (5 \times 100) = 500$

The number of boys who took admission in secondary school = $6n = (6 \times 100) = 600$

The number of girls who took admission in secondary school = $2m = (2 \times 100) = 200$

2013

The total number of students who took admission = 150% of $2000 = 3000$

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The total number of students who took admission in primary school = $3000/5 \times 3 = 1800$

The total number of students who took admission in secondary school = $3000/5 \times 2 = 1200$

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 3:4. Let it be $3p$ and $4p$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 1:3. Let it be q and $3q$ respectively.

Now,

$$3p + 3q = 1800$$

$$p + q = 600 \dots\dots\dots(V)$$

Also,

$$4p + q = 1200 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$12p + 12q = 7200$$

$$12p + 3q = 3600$$

$$9q = 3600$$

$$q = 400$$

Now, put the value of q in equation (V), we get,

$$p + q = 600$$

$$p + 400 = 600$$

$$p = 200$$

Therefore,

The number of boys who took admission in primary school = $3q = (3 \times 400) = 1200$

The number of girls who took admission in primary school = $3p = (3 \times 200) = 600$

The number of boys who took admission in secondary school = $4p = (4 \times 200) = 800$

The number of girls who took admission in secondary school = $q = 400$

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Year	The total number of students who took admission	The total number of students who took admission in primary school	The total number of students who took admission in secondary school	The number of boys who took admission in primary school	The number of girls who took admission in primary school	The number of boys who took admission in secondary school	The number of girls who took admission in secondary school
2011	1200	900	300	600	300	240	60
2012	2000	1200	800	700	500	600	200
2013	3000	1800	1200	1200	600	800	400
Total	6200	3900	2300	2500	1400	1640	660

Answer: B

Secondary School (2012)

The total number of girls who took admission = 200

The number of girls who did not appear in the examination = 10% of 200 = 20

The number of girls who appeared in the examination = $(200 - 20) = 180$

The number of girls who failed in the examination = 30

Therefore, the number of girls who passed in the examination in secondary school in 2012 = $(180 - 30) = 150$.

Hence, the correct answer is option B.

18. Questions

2011

The total number of students who took admission = 1200

The total number of students who took admission in secondary school = $1/4$ of 1200 = 300

The total number of students who took admission in primary school = $(1200 - 300) = 900$

Let the number of girls who took admission in primary school and boys who took admission in secondary school be $5a$ and $4a$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = $10:100 = 1:10$. Let it be b and $10b$ respectively.

Now,

$$5a + 10b = 900$$

$$a + 2b = 180 \dots\dots\dots(I)$$

And,

$$4a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$20a + 40b = 3600$$

$$20a + 5b = 1500$$

$$35b = 2100$$

$$b = 60$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 180$$

$$a + (2 * 60) = 180$$

$$a = 180 - 120$$

$$a = 60$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 10b = (10 * 60) = 600$$

$$\text{The number of girls who took admission in primary school} = 5a = (5 * 60) = 300$$

$$\text{The number of boys who took admission in secondary school} = 4a = (4 * 60) = 240$$

$$\text{The number of girls who took admission in secondary school} = b = 60$$

2012

The total number of students who took admission in primary school = $\frac{4}{3}$ of 900 = 1200 (**Note: 33.33% = $\frac{1}{3}$**)

The ratio of the total number of students who took admission in primary and secondary school = 3:2

$$\text{The total number of students who took admission in secondary school} = 1200/3*2 = 800$$

$$\text{The total number of students who took admission} = (1200 + 800) = 2000$$

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 2:7. Let it be 2m and 7m respectively.

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The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 5:6. Let it be 5n and 6n respectively. (**Note: 16.67% = $\frac{1}{6}$**)

Now,

$$7m + 5n = 1200 \dots\dots\dots(III)$$

And,

$$2m + 6n = 800$$

$$m + 3n = 400 \dots\dots\dots(IV)$$

By solving equations (III) and (IV), we get,

$$14m + 10n = 2400$$

$$7m + 5n = 1200$$

$$7m + 21n = 2800$$

$$16n = 1600$$

$$n = 100$$

Now, put the value of n in equation (III), we get,

$$7m + 5n = 1200$$

$$7m + (5 * 100) = 1200$$

$$7m = 1200 - 500$$

$$7m = 700$$

$$m = 100$$

Therefore,

The number of boys who took admission in primary school = $7m = (7 * 100) = 700$

The number of girls who took admission in primary school = $5n = (5 * 100) = 500$

The number of boys who took admission in secondary school = $6n = (6 * 100) = 600$

The number of girls who took admission in secondary school = $2m = (2 * 100) = 200$

2013

The total number of students who took admission = 150% of $2000 = 3000$

The total number of students who took admission in primary school = $3000/5*3 = 1800$

The total number of students who took admission in secondary school = $3000/5*2 = 1200$

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = $3:4$. Let it be $3p$ and $4p$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = $1:3$. Let it be q and $3q$ respectively.

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Now,

$$3p + 3q = 1800$$

$$p + q = 600 \dots \dots \dots (V)$$

Also,

$$4p + q = 1200 \dots \dots \dots (VI)$$

By solving equations (V) and (VI), we get,

$$12p + 12q = 7200$$

$$12p + 3q = 3600$$

$$9q = 3600$$

$$q = 400$$

Now, put the value of q in equation (V), we get,

$$p + q = 600$$

$$p + 400 = 600$$

$$p = 200$$

Therefore,

The number of boys who took admission in primary school = $3q = (3 * 400) = 1200$

The number of girls who took admission in primary school = $3p = (3 * 200) = 600$

The number of boys who took admission in secondary school = $4p = (4 * 200) = 800$

The number of girls who took admission in secondary school = $q = 400$

Year	The total number of students who took admission	The total number of students who took admission in primary school	The total number of students who took admission in secondary school	The number of boys who took admission in primary school	The number of girls who took admission in primary school	The number of boys who took admission in secondary school	The number of girls who took admission in secondary school
2011	1200	900	300	600	300	240	60
2012	2000	1200	800	700	500	600	200
2013	3000	1800	1200	1200	600	800	400
Total	6200	3900	2300	2500	1400	1640	660

Answer: C

2014

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The number of students who took admission in secondary school = 800

The number of students who took admission in primary school = 1600

So, the total number of students who took admission = $(800 + 1600) = 2400$

Therefore, ? = 2400.

Hence, the correct answer is option C.

19. Questions

2011

The total number of students who took admission = 1200

The total number of students who took admission in secondary school = $1/4$ of 1200 = 300

The total number of students who took admission in primary school = $(1200 - 300) = 900$

Let the number of girls who took admission in primary school and boys who took admission in secondary school be $5a$ and $4a$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = $10:100 = 1:10$. Let it be b and $10b$ respectively.

Now,

$$5a + 10b = 900$$

$$a + 2b = 180 \dots\dots\dots(I)$$

And,

$$4a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$20a + 40b = 3600$$

$$20a + 5b = 1500$$

$$35b = 2100$$

$$b = 60$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 180$$

$$a + (2 * 60) = 180$$

$$a = 180 - 120$$

$$a = 60$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 10b = (10 * 60) = 600$$

$$\text{The number of girls who took admission in primary school} = 5a = (5 * 60) = 300$$

$$\text{The number of boys who took admission in secondary school} = 4a = (4 * 60) = 240$$

$$\text{The number of girls who took admission in secondary school} = b = 60$$

2012

The total number of students who took admission in primary school = $4/3$ of 900 = 1200 (**Note: $33.33\% = 1/3$**)

The ratio of the total number of students who took admission in primary and secondary school = 3:2

The total number of students who took admission in secondary school = $1200/3 \times 2 = 800$

The total number of students who took admission = $(1200 + 800) = 2000$

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 2:7. Let it be $2m$ and $7m$ respectively.

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 5:6. Let it be $5n$ and $6n$ respectively. (**Note: $16.67\% = 1/6$**)

Now,

$$7m + 5n = 1200 \dots \dots \dots (III)$$

And,

$$2m + 6n = 800$$

$$m + 3n = 400 \dots \dots \dots (IV)$$

By solving equations (III) and (IV), we get,

$$14m + 10n = 2400$$

$$7m + 5n = 1200$$

$$7m + 21n = 2800$$

$$16n = 1600$$

$$n = 100$$

Now, put the value of n in equation (III), we get,

$$7m + 5n = 1200$$

$$7m + (5 \times 100) = 1200$$

$$7m = 1200 - 500$$

$$7m = 700$$

$$m = 100$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 7m = (7 \times 100) = 700$$

$$\text{The number of girls who took admission in primary school} = 5n = (5 \times 100) = 500$$

$$\text{The number of boys who took admission in secondary school} = 6n = (6 \times 100) = 600$$

$$\text{The number of girls who took admission in secondary school} = 2m = (2 \times 100) = 200$$

2013

$$\text{The total number of students who took admission} = 150\% \text{ of } 2000 = 3000$$

$$\text{The total number of students who took admission in primary school} = 3000/5 \times 3 = 1800$$

$$\text{The total number of students who took admission in secondary school} = 3000/5 \times 2 = 1200$$

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 3:4. Let it be $3p$ and $4p$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 1:3. Let it be q and $3q$ respectively.

Now,

$$3p + 3q = 1800$$

$$p + q = 600 \dots\dots\dots(V)$$

Also,

$$4p + q = 1200 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$12p + 12q = 7200$$

$$12p + 3q = 3600$$

$$9q = 3600$$

$$q = 400$$

Now, put the value of q in equation (V), we get,

$$p + q = 600$$

$$p + 400 = 600$$

$$p = 200$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 3q = (3 * 400) = 1200$$

$$\text{The number of girls who took admission in primary school} = 3p = (3 * 200) = 600$$

$$\text{The number of boys who took admission in secondary school} = 4p = (4 * 200) = 800$$

$$\text{The number of girls who took admission in secondary school} = q = 400$$

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Year	The total number of students who took admission	The total number of students who took admission in primary school	The total number of students who took admission in secondary school	The number of boys who took admission in primary school	The number of girls who took admission in primary school	The number of boys who took admission in secondary school	The number of girls who took admission in secondary school
2011	1200	900	300	600	300	240	60
2012	2000	1200	800	700	500	600	200
2013	3000	1800	1200	1200	600	800	400
Total	6200	3900	2300	2500	1400	1640	660

Answer: E

The number of boys who took admission in primary school in 2012 = 700

The number of boys who took admission in secondary school in 2013 = 800

Therefore, the required sum = $(700 + 800) = 1500$

Hence, the correct answer is option E.

20. Questions

2011

The total number of students who took admission = 1200

The total number of students who took admission in secondary school = $\frac{1}{4}$ of 1200 = 300

The total number of students who took admission in primary school = $(1200 - 300) = 900$

Let the number of girls who took admission in primary school and boys who took admission in secondary school be $5a$ and $4a$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = $10:100 = 1:10$. Let it be b and $10b$ respectively.

Now,

$$5a + 10b = 900$$

$$a + 2b = 180 \dots\dots\dots(I)$$

And,

$$4a + b = 300 \dots\dots\dots(II)$$

By solving equations (I) and (II), we get,

$$20a + 40b = 3600$$

$$20a + 5b = 1500$$

$$35b = 2100$$

$$b = 60$$

Now, put the value of b in equation (I), we get,

$$a + 2b = 180$$

$$a + (2 * 60) = 180$$

$$a = 180 - 120$$

$$a = 60$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 10b = (10 * 60) = 600$$

$$\text{The number of girls who took admission in primary school} = 5a = (5 * 60) = 300$$

$$\text{The number of boys who took admission in secondary school} = 4a = (4 * 60) = 240$$

$$\text{The number of girls who took admission in secondary school} = b = 60$$

2012

The total number of students who took admission in primary school = $\frac{4}{3}$ of 900 = 1200 (**Note: 33.33% = $\frac{1}{3}$**)

The ratio of the total number of students who took admission in primary and secondary school = 3:2

$$\text{The total number of students who took admission in secondary school} = \frac{1200}{3} * 2 = 800$$

$$\text{The total number of students who took admission} = (1200 + 800) = 2000$$

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 2:7. Let it be 2m and 7m respectively.

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 5:6. Let it be 5n and 6n respectively. (**Note: 16.67% = $\frac{1}{6}$**)

Now,

$$7m + 5n = 1200 \dots \dots \dots \text{(III)}$$

And,

$$2m + 6n = 800$$

$$m + 3n = 400 \dots \dots \dots \text{(IV)}$$

By solving equations (III) and (IV), we get,

$$14m + 10n = 2400$$

$$7m + 5n = 1200$$

$$7m + 21n = 2800$$

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$$16n = 1600$$

$$n = 100$$

Now, put the value of n in equation (III), we get,

$$7m + 5n = 1200$$

$$7m + (5 * 100) = 1200$$

$$7m = 1200 - 500$$

$$7m = 700$$

$$m = 100$$

Therefore,

$$\text{The number of boys who took admission in primary school} = 7m = (7 * 100) = 700$$

$$\text{The number of girls who took admission in primary school} = 5n = (5 * 100) = 500$$

$$\text{The number of boys who took admission in secondary school} = 6n = (6 * 100) = 600$$

$$\text{The number of girls who took admission in secondary school} = 2m = (2 * 100) = 200$$

2013

$$\text{The total number of students who took admission} = 150\% \text{ of } 2000 = 3000$$

$$\text{The total number of students who took admission in primary school} = 3000/5*3 = 1800$$

$$\text{The total number of students who took admission in secondary school} = 3000/5*2 = 1200$$

The ratio of the number of girls who took admission in primary school and boys who took admission in secondary school = 3:4. Let it be $3p$ and $4p$ respectively.

The ratio of the number of girls who took admission in secondary school and boys who took admission in primary school = 1:3. Let it be q and $3q$ respectively.

Now,

$$3p + 3q = 1800$$

$$p + q = 600 \dots\dots\dots(V)$$

Also,

$$4p + q = 1200 \dots\dots\dots(VI)$$

By solving equations (V) and (VI), we get,

$$12p + 12q = 7200$$

$$12p + 3q = 3600$$

$$9q = 3600$$

$$q = 400$$

Now, put the value of q in equation (V), we get,

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$$p + q = 600$$

$$p + 400 = 600$$

$$p = 200$$

Therefore,

The number of boys who took admission in primary school = $3q = (3 * 400) = 1200$

The number of girls who took admission in primary school = $3p = (3 * 200) = 600$

The number of boys who took admission in secondary school = $4p = (4 * 200) = 800$

The number of girls who took admission in secondary school = $q = 400$

Year	The total number of students who took admission	The total number of students who took admission in primary school	The total number of students who took admission in secondary school	The number of boys who took admission in primary school	The number of girls who took admission in primary school	The number of boys who took admission in secondary school	The number of girls who took admission in secondary school
2011	1200	900	300	600	300	240	60
2012	2000	1200	800	700	500	600	200
2013	3000	1800	1200	1200	600	800	400
Total	6200	3900	2300	2500	1400	1640	660

Answer: B

2011

Secondary School

The total number of girls who took admission = 60

The number of girls who use own transport = $1/6$ of 60 = 10

The number of girls who use school transport = $(60 - 10) = 50$

Primary

The total number of girls who took admission = 300

The number of girls who use own transport = $10/10 * 100 = 100$

The number of girls who use school transport = $(300 - 100) = 200$

Therefore, the required difference = $(200 - 50) = 150$.

Hence, the correct answer is option B.

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